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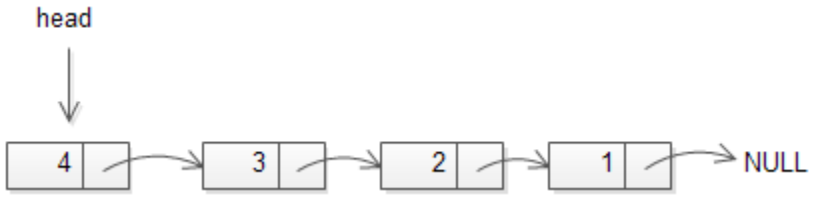
Data Structures and Algorithms – ICT2113 **Level 2- Semester 1**

Laboratory Assignment 4

Objectives

Main objective of this lab session is to understand the concept of Linked list and Linked list based implementation of List.

- A linked list is a dynamic data structure
- A linked list grow and shrink during their life time.
- Link list elements are not physically adjacent in memory



1. Write a C program to create and display a Singly Linked List.
   Test Data :

   Input the number of nodes: 3
   Input data for node 1 : 15
   Input data for node 2 : 9
   Input data for node 3 : 12

   Output:
   Data entered in the list:
   Data =15
   Data = 9
   Data = 12

   Create a node with two members

```
struct node
{
        int num;
        struct node *nextptr;
}*stnode;
```

void createNodeList(int n); // function to create the list
void displayList(); // function to display the list

2. Write a C program to maintain a Linked list based ordered list.
    a. Add three elements to the list. Implement the ***createNodeList()*** function.
    b. Display the list node. Implement the ***displayList()*** function.
    c. Search the given node from the list. Implement the ***FindElement(int)*** function.
    d. Insert a node at the beginning of the linked list. Implement the ***NodeInsertatBegin(int)*** function.
    e. Display the list.
    f. Insert a node at the end of the linked list. Implement the function ***NodeInsertatEnd(int)*** function.
    g. Display the list.
    h. Count the number of nodes of the linked link. Implement the ***int NodeCount()***function.
    i. Insert a node at middle of the list. Implement the ***insertNodeAtMiddle(int num, int pos)*** function.
    j. Display the list.
    k. Delete the first node of the linked list. Implement the ***FirstNodeDeletion()***function.
    l. Display the list.
    m. Count the number of nodes of the linked link.

3. Write a program in C to create and display a doubly linked list.
   Function's prototype are:

   void DlListcreation(int);
   void displayDlList();